SOFTWARE REQUIREMENTS SPECIFICATION

I-Property: Rental Property Management System

Version 3.0 | Specification Grade | Based on BRD v1.3

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1. Related Document Register

This SRS is one artifact within a wider project document set. Each companion document is referenced in the relevant section of this SRS so a reader can locate the authoritative source for any topic.

Doc ID	Title	Ver	Status	Owner	SRS Sections That Reference It
BRD-001	Business Requirements Document	1.3	Approved	Admin / IT-PMO	All — parent document
SAD-001	Software Architecture Document	TBD	Pending	IT Architecture	§5 — Architecture Overview
SDD-001	Software Design Document	TBD	Pending	Development Lead	§6 (processing logic), §10 (field definitions)
UI-001	UI/UX Design Specification & Wireframes	TBD	Pending	UX Designer	§9 — Screen Inventory & UI Requirements
IDD-001	Integration Design Document (OF)	TBD	Pending	Integration Architect	§8 — API & Integration Specifications
TP-001	Test Plan & Test Cases	TBD	Pending	QA Lead	§13 — RTM; Verification Steps in §6 & §7
DP-001	Deployment & Infrastructure Plan	TBD	Pending	IT Infrastructure	§11 — NFR verification methods
TRM-001	Full Requirements Traceability Matrix	TBD	Pending	BA / PMO	§13 — RTM preview

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3. Introduction

3.1 Purpose

This document specifies the software requirements for the I-Property system at a level of detail sufficient for a development team to implement, a QA team to write test cases, and stakeholders to verify that the delivered system meets the stated business needs. Every functional requirement includes exact inputs, processing logic, outputs, error conditions, and step-by-step verification procedures.

3.2 Intended Audience

Audience	How They Use This Document
Solution Architects	Sections 4, 7, 10 — architecture decisions, API contracts, NFRs
Backend Developers	Section 5 — FR processing logic, error codes; Section 7 — API specs; Section 9 — field definitions
Frontend Developers	Section 8 — screen inventory and UI requirements; Section 9 — field validation rules
QA / Test Engineers	Verification Steps within each FR block (Section 5 & 6); Section 12 — RTM
Business Stakeholders	Sections 5, 6 — requirement descriptions and expected outputs for sign-off validation
Integration Team	Section 7 — Oracle Financial API contracts, payload schemas, error handling
Database Architects	Section 9 — field definitions and data types (elaborated in SDD-001 §4)

3.3 Scope

- Phase 1: New request, renewal, and vacated/termination workflows with full multi-level approval routing
- Phase 1: Utility bills payment tracking and ABH approval
- Phase 1: Automated monthly payment summaries, Oracle Financial integration, notification engine
- Phase 1: Role-based access control (7 roles), audit trail, document management
- Phase 2 (out of scope for this SRS): Signage module, Capex integration, cheque cancellation
- Out of scope always: Payment disbursement execution (handled entirely within Oracle Financial)

3.4 Conventions

- All timestamps are stored in UTC; displayed in Pacific Time Zone (UTC -8) in the UI
- All monetary amounts are in USD unless stated otherwise
- API base URL: https://iproperty.jvo.internal/api/v1 (to be confirmed in IDD-001)
- HTTP status codes follow RFC 7231
- All API requests require Authorization: Bearer {JWT} header
- Field lengths refer to string character counts or numeric precision (scale,precision)

4. System Overview

4.1 System Context

I-Property is a multi-role web application deployed on J-Vo's internal infrastructure. It does not execute payments. It generates payment requests and transmits them to Oracle Financial via INT-01 (see Section 7). The full system context diagram is in SAD-001 S3.

4.2 User Roles & Permissions Summary

Role	System Access	Cannot Do
Area Officer	Create new/renewal/vacated requests; enter utility bills; view own requests	Approve any request; view other areas; access Finance screens
Area Admin	All Area Officer permissions; access Print Queue; initiate vacated requests	Approve requests; view payment amounts
ABH / MR / ARH	Approve/reject requests; approve monthly payment summaries; stop payments; view regional reports	Create requests; access User Management
Director Office	Approve/reject requests escalated from ABH; view request history	Create requests; view payment data
Admin Head Office	Review and complete requests; edit agreement details; upload vacated notices	Approve as ABH or Director; manage users
Finance Dept.	View and export payment reports; mark stop-payment notes; read-only access to agreements	Create or approve requests; edit agreement data
System Admin	Full access: user management, role assignment, notification config, audit log viewer	Approve business requests (no business role)

5. System Architecture Overview

Full architectural specification is in SAD-001. This section provides the context required to understand the system behaviour described in later sections.

5.1 Tier Structure

Tier	Technology (to be confirmed in SAD-001)	Key Responsibility
Presentation	React SPA served over HTTPS; responsive desktop-first layout defined in UI-001	All screens listed in Section 8; form validation (client-side, duplicate of server-side)
Application	RESTful API layer (Node.js / Java Spring — TBD in SAD-001)	Business logic, workflow engine, approval routing, notification queue, OF integration adapters
Data	Relational RDBMS (PostgreSQL / Oracle — TBD in SAD-001)	All entities in Section 9; audit log table (append-only); document path references
Document Store	Network file share or object store (S3-compatible — TBD in DP-001)	Binary storage for uploaded files; application stores path reference in DB only

Notification	Async job queue (Redis-backed — TBD in SAD-001) feeding SMTP relay	Decoupled notification delivery; retry on failure; prevents blocking of approval transactions

5.2 Authentication & Session

- Authentication via organization SSO/LDAP using OAuth 2.0 / SAML 2.0 (to be confirmed in SAD-001 S5)
- JWT issued on successful login; expiry: 8 hours; refresh token: 24 hours
- Session inactivity timeout: 30 minutes; UI shows 2-minute warning modal before auto-logout
- Account lockout after 5 consecutive failed login attempts; unlock by System Admin or 30-minute auto-reset
- All API endpoints validate JWT on every request; 401 returned if token absent, expired, or invalid

5.3 Audit Architecture

- Separate audit_log table: columns: log_id, table_name, record_id, action (INSERT/UPDATE/DELETE), old_value (JSON), new_value (JSON), user_id, ip_address, timestamp_utc
- Triggers or application-layer interceptors write to audit_log on every CUD operation
- No UPDATE or DELETE permitted on audit_log table; enforced at database role level
- Audit logs retained 3 years; archived to cold storage thereafter; readable via Audit Log Viewer (System Admin only)

6. Functional Requirements

Each requirement is specified with: description, pre-conditions, trigger, precise inputs, processing logic (including business rules and calculations), expected outputs, error handling with specific error codes and messages, and step-by-step verification procedures for QA.

6.1 Request Initiation & Reference ID Generation

FR-NR-01 — Request Submission & Reference ID Generation [Must Have] | BRD: FR-NR-01, FR-NR-02	
Description	Area Officer completes and submits a new rental request form. The system validates all inputs, persists the record, generates a globally unique reference ID, and routes the request to the appropriate ABH based on the area selected.
Pre-conditions	User is authenticated with role = Area Officer User's assigned area is configured in the system At least one active Landlord Profile and one active Property Profile exist
Trigger	User clicks the Submit button on the New Rental Request form (SCR-03)
Inputs	landlord_id: string, required — must match active record in landlord_profile table property_id: string, required — must match active record in property_profile table linked to landlord_id area_code: string, required — must match a configured area in areas table; must match user's assigned area agreement_start_date: date (DD/MM/YYYY), required — must not be in the past agreement_end_date: date (DD/MM/YYYY), required — must be > agreement_start_date rental_amount: decimal(15,2), required — range: 1.00 to 99,999,999.99

	tax_rate: decimal(5,2), required — range: 0.00 to 100.00 agreement_type: enum [New, Renewal, Transfer], required document_attachments: file[], optional — PDF/JPG/PNG; max 10 MB per file; max 5 files
Processing Logic	1. Server-side validation executes for all fields (client-side validation is a UX duplicate only) 2. If any validation fails: return HTTP 422 with error array — do NOT persist any data 3. If all valid: open DB transaction 4. Insert record into requests table with status = PENDING_ABH 5. Generate reference_id: format REQ-{YYYY}-{6-digit zero-padded sequence} e.g. REQ-2025-000142. Sequence resets each calendar year. Use DB sequence object to guarantee uniqueness under concurrent submissions 6. Look up ABH user ID(s) for the submitted area_code from area_abh_mapping table 7. Commit transaction 8. Enqueue notification job NE-01 (async — must not block HTTP response) 9. Return HTTP 201 response
Outputs / Response	HTTP 201 Created Body: { request_id: 'REQ-2025-000142', status: 'PENDING_ABH', created_at: '2025-04-01T08:32:11Z', assigned_abh_ids: ['USR-0045'] } UI: Displays success banner with reference ID; form is cleared and locked DB: New row in requests table; new row in request_audit_trail table Email: NE-01 dispatched to ABH within 2 minutes (see Section 11)
Error Handling	ERR-NR-01-001: landlord_id not found → HTTP 422, field: landlord_id, message: 'Landlord ID does not exist or is inactive' ERR-NR-01-002: property_id not linked to landlord_id → HTTP 422, field: property_id, message: 'Property is not associated with the selected Landlord' ERR-NR-01-003: area_code not in user's allowed areas → HTTP 403, message: 'You do not have permission to submit requests for this area' ERR-NR-01-004: agreement_start_date in past → HTTP 422, field: agreement_start_date, message: 'Start date cannot be in the past' ERR-NR-01-005: agreement_end_date <= start_date → HTTP 422, field: agreement_end_date, message: 'End date must be after Start date' ERR-NR-01-006: rental_amount out of range → HTTP 422, field: rental_amount, message: 'Amount must be between 1.00 and 99,999,999.99' ERR-NR-01-007: file type not allowed → HTTP 422, field: document_attachments, message: 'Only PDF, JPG, and PNG files are permitted' ERR-NR-01-008: file size > 10 MB → HTTP 422, field: document_attachments, message: 'File {filename} exceeds the 10 MB limit' ERR-NR-01-009: DB transaction failure → HTTP 500, message: 'Request could not be saved. Please try again. If the issue persists, contact IT Support.' Log full error server-side; do not expose stack trace to client
Verification Steps	TC-001-A: Submit fully valid request → verify HTTP 201, reference ID format REQ-YYYY-NNNNNN, record in DB with status=PENDING_ABH, NE-01 email received by ABH within 2 minutes TC-001-B: Submit with invalid landlord_id → verify HTTP 422, correct error code ERR-NR-01-001, no DB record created TC-001-C: Submit with start_date = yesterday → verify HTTP 422, ERR-NR-01-004 TC-001-D: Simulate 50 concurrent submissions → verify all receive unique reference IDs with no duplicates (concurrency test) TC-001-E: Submit with 11 MB file → verify HTTP 422, ERR-NR-01-008, file not stored on disk

6.2 Approval Routing & Status Transitions

FR-NR-02 — ABH Approve or Reject [Must Have] | BRD: FR-NR-03, FR-NR-04	
Description	ABH reviews a pending request and takes an Approve or Reject action. The system enforces mandatory remarks on rejection, transitions the request status, and routes or closes accordingly.

Pre-conditions	Request exists with status = PENDING_ABH Authenticated user has role ABH/MR/ARH and is mapped to the request's area_code
Trigger	ABH clicks Approve or Reject on the Approval Queue screen (SCR-04) or Request Detail screen (SCR-05)
Inputs	request_id: string, required action: enum [APPROVE, REJECT], required remarks: string, required if action = REJECT — min 1 char, max 500 chars
Processing Logic	1. Validate: request status must be PENDING_ABH; if not → ERR-AP-02-001 2. Validate: acting user must be mapped ABH for request's area; if not → HTTP 403 3. If action = APPROVE: update requests.status = PENDING_DIRECTOR; insert row in approval_history (approver_id, role, action=APPROVED, remarks=null, timestamp) 4. If action = REJECT: validate remarks not empty → ERR-AP-02-002; update requests.status = REJECTED; insert row in approval_history (action=REJECTED, remarks); no further routing 5. Commit transaction; enqueue appropriate notification (NE-02 for approve, NE-03 for reject)
Outputs / Response	HTTP 200 OK Body (approve): { request_id, new_status: 'PENDING_DIRECTOR', actioned_by: 'USR-0045', timestamp: '...' } Body (reject): { request_id, new_status: 'REJECTED', remarks: '...' } UI: Request removed from ABH's Approval Queue; status badge updated on Request Detail
Error Handling	ERR-AP-02-001: Request not in PENDING_ABH status → HTTP 409, message: 'This request has already been actioned (current status: {status})' ERR-AP-02-002: Reject attempted without remarks → HTTP 422, field: remarks, message: 'Rejection remarks are mandatory' ERR-AP-02-003: Acting user not mapped to request area → HTTP 403, message: 'You are not authorised to action requests for this area'
Verification Steps	TC-002-A: Approve valid PENDING_ABH request → verify status = PENDING_DIRECTOR, approval_history row, NE-02 email to Director within 2 min TC-002-B: Reject without remarks → verify HTTP 422, ERR-AP-02-002, status unchanged in DB TC-002-C: Reject with valid remarks → verify status = REJECTED, approval_history row with remarks, NE-03 email to Area Officer TC-002-D: ABH from Area B attempts to action Area A request → verify HTTP 403 TC-002-E: Attempt to approve already-approved request (status = PENDING_DIRECTOR) → verify HTTP 409 ERR-AP-02-001

FR-NR-03 — Director & Admin HO Approval — Full Chain Completion [Must Have] | BRD: FR-NR-04, FR-NR-05, FR-NR-06	
Description	Director Office and Admin Head Office complete the remaining approval chain. Admin HO marks the request Completed, which triggers simultaneous notifications to Area Admin and Finance.
Pre-conditions	For Director action: request status = PENDING_DIRECTOR; user role = Director Office For Admin HO action: request status = PENDING_ADMIN_HO; user role = Admin Head Office
Trigger	Approver clicks Approve / Reject on SCR-04 or SCR-05; Admin HO clicks Mark as Completed
Inputs	request_id, action (APPROVE/REJECT/COMPLETE), remarks (mandatory on REJECT)
Processing Logic	Director APPROVE: status → PENDING_ADMIN_HO; NE-04 sent to Admin HO Director REJECT: status → REJECTED; NE-05 sent to ABH + Area Officer Admin HO COMPLETE: status → COMPLETED; generate Approval Summary PDF (see FR-NR-04); enqueue NE-06 to Area Admin AND Finance simultaneously All transitions: insert row in approval_history; commit before enqueuing notifications

Outputs / Response	HTTP 200 with new_status in body for each action On COMPLETE: approval_summary_url returned in response body pointing to generated PDF
Error Handling	Same error pattern as FR-NR-02 with status-appropriate codes ERR-AP-03-001: Admin HO attempts Complete when status != PENDING_ADMIN_HO → HTTP 409
Verification Steps	TC-003-A: Full happy path — submit → ABH approve → Director approve → Admin HO complete → verify all status transitions in DB, all 4 notifications received, approval_summary_url accessible TC-003-B: Director rejects → verify NE-05 sent to both ABH and Area Officer (not just one) TC-003-C: Finance notification body must contain: agreement_id, landlord_name, rental_amount, tax_rate, effective_date — verify email content against NE-06 template fields

6.3 Approval Summary PDF Generation

FR-NR-04 — Auto-Generate Approval Summary PDF [Must Have] | BRD: FR-NR-08	
Description	When Admin HO marks a request as Completed, the system auto-generates an Approval Summary PDF that lists every approver in chain order with name, role, action taken, and timestamp. This PDF is appended to the print package.
Pre-conditions	Request status has just transitioned to COMPLETED All approval_history rows for the request exist in DB
Trigger	Triggered automatically by the COMPLETE action in FR-NR-03 (not a user-initiated action)
Inputs	Internally derived from approval_history table: approver_name, role, action, remarks (if any), timestamp_utc converted to PST
Processing Logic	1. Query approval_history WHERE request_id = X ORDER BY timestamp ASC 2. Render PDF template with: request reference_id, property address, landlord name, table of approvers (sequence, name, role, action, date/time PST, remarks) 3. Store PDF in document store; insert reference in request_documents table with doc_type = APPROVAL_SUMMARY 4. Return document URL in the COMPLETE response body 5. PDF must be generated within 10 seconds of the COMPLETE action
Outputs / Response	PDF file accessible at returned URL; filename format: APPROVAL_SUMMARY_{request_id}.pdf PDF contains: header (J-Vo logo placeholder, document title, reference ID), approver table with all chain members, footer with generation timestamp Document listed in Print Queue (SCR-11) alongside agreement documents
Error Handling	ERR-PDF-04-001: PDF generation timeout > 10 seconds → HTTP 202 Accepted (not 200); response body includes { pdf_status: 'GENERATING', poll_url: '/api/v1/requests/{id}/pdf-status' }; UI shows spinner; polls every 2 seconds ERR-PDF-04-002: PDF generation failure → log error; set pdf_status = FAILED; System Admin alerted; Area Admin shown: 'Summary PDF generation failed. Contact IT Support (Ref: {log_id})'
Verification Steps	TC-004-A: Complete a request with 3-level approval chain → verify PDF generated within 10s, PDF contains all 3 approvers with correct names/timestamps, PDF accessible via returned URL TC-004-B: Verify PDF timestamp is in PST (UTC-8) TC-004-C: Simulate PDF engine timeout → verify HTTP 202, poll_url in response, UI shows generating state, PDF eventually available

6.4 Form Abandonment — Unsaved Changes

FR-NR-05 — Discard Unsaved Changes on Navigation Away [Must Have] | BRD: FR-NR-12	
Description	If a user navigates away from a partially completed form without saving, the system prompts for confirmation. If confirmed, all entered data is discarded. No partial record is persisted to the database.

Pre-conditions	User has opened any create/edit form (SCR-03, SCR-07, SCR-08) User has modified at least one field without submitting
Trigger	User triggers browser Back, closes tab, navigates to another screen via menu, or clicks Cancel
Inputs	Navigation intent event (browser before unload or React Router navigation guard)
Processing Logic	1. Intercept navigation using browser before unload event and in-app router guard 2. If the form is dirty (any field changed from its initial state): display modal — title: 'Unsaved Changes', body: 'You have unsaved changes. If you leave, your changes will be lost.', buttons: [Stay on Page] [Discard Changes] 3. If user clicks Stay on Page: dismiss modal, user remains on current form — no data loss 4. If user clicks Discard Changes: clear form state, reset to initial values, proceed with navigation 5. If form has NOT been modified: allow navigation without prompt 6. On browser tab close: browser's native beforeunload fires — custom message may not display in all browsers; this is an accepted limitation
Outputs / Response	Modal displayed within 100ms of navigation event No data written to DB at any point during form fill until Submit is clicked On discard: form resets to empty (new form) or original values (edit form)
Error Handling	No specific error codes — failure mode is: modal does not appear, risking accidental data loss Mitigation: automated regression test in TP-001 must cover this scenario in Chrome, Firefox, and Edge
Verification Steps	TC-005-A: Fill 3 fields on New Request form, click browser Back → verify modal appears TC-005-B: Fill fields, click Discard → verify form clears, no DB record exists for that session TC-005-C: Fill fields, click Stay on Page → verify all entered data is preserved TC-005-D: Open form, do NOT modify any field, click Back → verify no modal (form not dirty) TC-005-E: Repeat TC-005-A in Firefox and Edge

6.5 Monthly Payment Summary

FR-NR-06 — Auto-Generate & Approve Monthly Payment Summary [Must Have] | BRD: FR-NR-09, FR-NR-10	
Description	On the last working day of each month, the system auto-generates a payment summary covering all active agreements. ABH/GM reviews and approves it with the option to stop payment for specific landlords. Finance receives the approved report.
Pre-conditions	Active agreements exist with status = COMPLETED and payment not stopped System scheduler is operational; working day calendar is configured
Trigger	Automated scheduler trigger at 08:00 PST on last working day of month
Inputs	Internally derived: all active agreements, rental amounts, tax rates, bank details ABH input: stop_payment flag (boolean), stop_reason (string, min 10 chars if flag = true)
Processing Logic	Scheduler: 1. Query all agreements WHERE status=ACTIVE AND payment_stopped=false 2. For each: calculate gross_amount, tax_deduction (rental_amount * tax_rate / 100), net_payable = gross_amount - tax_deduction 3. Insert batch into monthly_payment_summary table with period = current month, summary_status = PENDING_ABH 4. Enqueue NE-07 to all ABH/GM users ABH review: 5. ABH may set payment_stopped=true on any row with mandatory stop_reason 6. ABH clicks Approve Summary → summary_status = APPROVED; enqueue NE-08 to Finance 7. Report locked — no further edits permitted once status = APPROVED

Outputs / Response	Scheduler: NE-07 email to ABH/GM by 08:05 PST on generation date Finance receives NE-08 email with download link to approved report Report exportable as Excel (.xlsx) and PDF; Excel contains one row per landlord with all payment fields Stopped payments appear in report with status = STOPPED and stop_reason column populated
Error Handling	ERR-PS-06-001: Scheduler fails to run → System Admin receives automated alert; manual trigger endpoint available: POST /api/v1/payment-summary/generate with admin auth ERR-PS-06-002: Stop payment attempted without stop_reason → HTTP 422, field: stop_reason, message: 'A reason is required to stop a payment (minimum 10 characters)' ERR-PS-06-003: Attempt to edit approved summary → HTTP 409, message: 'This summary has been approved and is locked. Raise a Change Request to amend.'
Verification Steps	TC-006-A: Trigger scheduler (via admin endpoint in test env) → verify summary generated for all active agreements, NE-07 email received TC-006-B: Set stop_payment without reason → verify HTTP 422 ERR-PS-06-002 TC-006-C: Set stop_payment with valid reason → verify payment_stopped=true in DB, stopped entry in Excel export with reason TC-006-D: Approve summary → verify summary_status=APPROVED, NE-08 sent to Finance, subsequent edit attempt returns HTTP 409 TC-006-E: Verify Excel export — each row must contain: landlord_id, landlord_name, agreement_id, period, gross_amount, tax_deduction, net_payable, bank_account (masked: last 4 digits visible), status

6.6 Vacated / Termination — Stop Payment

FR-VT-01 — Automated Stop-Payment on Vacated Approval [Must Have] | BRD: FR-VT-01, FR-VT-02	
Description	When the Director approves a vacated/termination request, the system immediately halts payment generation for that agreement and notifies Finance with precise stop-payment details. No manual intervention should be required.
Pre-conditions	Vacated request exists with status = PENDING_DIRECTOR Agreement linked to the request has status = ACTIVE
Trigger	Director clicks Approve on a vacated request
Inputs	request_id, action=APPROVE, termination_date (captured at request creation)
Processing Logic	1. Update requests.status = PENDING_ADMIN_HO (standard chain) AND set agreements.payment_stopped = true, agreements.payment_stop_date = termination_date, agreements.stop_reason = 'VACATED' 2. Remove agreement from future payment scheduler runs immediately (scheduler checks payment_stopped flag) 3. Enqueue NE-10 (urgent) to Finance with: agreement_id, landlord_id, landlord_name, last_payment_date = termination_date - 1 month (if mid-cycle) or = termination_date, termination_date 4. Insert audit_log entry: action=PAYMENT_STOPPED, triggered_by=vacated_approval, request_id 5. All above steps in a single DB transaction — if any step fails, rollback all
Outputs / Response	HTTP 200; Finance NE-10 email dispatched within 2 minutes Agreement excluded from next monthly_payment_summary generation run Audit log entry verifiable by System Admin
Error Handling	ERR-VT-01-001: DB transaction rollback → HTTP 500; Finance NOT notified (notification only sent on successful commit); System Admin alerted; manual intervention required — documented in DP-001 §6 If Finance notification delivery fails (SMTP error): notification_log records status=FAILED; retry 3 times at 5-minute intervals; if still failed: System Admin alerted with log_id
Verification Steps	TC-007-A: Approve vacated request → verify agreement.payment_stopped=true in DB, payment_stop_date = termination_date, NE-10 email received by Finance within 2 min

	TC-007-B: Run payment scheduler after TC-007-A → verify vacated agreement NOT included in generated summary TC-007-C: Verify NE-10 email body contains all required fields: agreement_id, landlord_id, landlord_name, last_payment_date, termination_date TC-007-D: Simulate DB failure during transaction → verify rollback, agreement.payment_stopped=false, Finance NOT notified, System Admin alert triggered

6.7 Utility Bill Submission & Duplicate Prevention

FR-UT-01 — Utility Bill Entry with Duplicate Check [Must Have] | BRD: FR-UT-01 to FR-UT-04	
Description	The Area Officer submits a utility bill against an active agreement. The system validates all inputs, checks for duplicate entries (same agreement + billing period + bill type), routes to ABH for approval, and notifies Finance upon ABH approval.
Pre-conditions	Agreement with utility connections exists in agreements table with status = ACTIVE User has role = Area Officer with access to the agreement's area
Trigger	User completes Utility Bill form (SCR-08) and clicks Submit
Inputs	agreement_id: string, required — must reference ACTIVE agreement bill_type: enum [ELECTRICITY, GAS, WATER, OTHER], required billing_period: string (MM/YYYY), required — must not be in the future units_consumed: decimal(10,2), required — min 0 bill_amount: decimal(15,2), required — min 0.01 advance_tax: decimal(15,2), optional — default 0, min 0 bill_document: file, required — PDF/JPG/PNG, max 10 MB
Processing Logic	1. Validate all fields per rules above 2. Duplicate check: SELECT COUNT(*) FROM utility_bills WHERE agreement_id=X AND bill_type=Y AND billing_period=Z AND status != 'REJECTED' → if count > 0: ERR-UT-01-003 3. Calculate net_payable = bill_amount + advance_tax (stored as computed/derived field) 4. Store uploaded file in document store; record path in utility_bills.document_path 5. Insert utility_bills row with status = PENDING_ABH 6. Enqueue NE-11 to ABH for the agreement's area
Outputs / Response	HTTP 201 Created Body: { utility_bill_id: 'UTIL-2025-000089', status: 'PENDING_ABH', net_payable: 12500.00 } NE-11 email sent to ABH within 2 minutes
Error Handling	ERR-UT-01-001: agreement_id not ACTIVE → HTTP 422, message: 'Agreement is not active or does not exist' ERR-UT-01-002: billing_period in future → HTTP 422, message: 'Billing period cannot be in the future' ERR-UT-01-003: Duplicate bill → HTTP 409, message: 'A bill for {bill_type} / {billing_period} already exists for this agreement (ID: {existing_bill_id}). To correct it, contact your Admin.' ERR-UT-01-004: bill_amount = 0 → HTTP 422, message: 'Bill amount must be greater than zero'
Verification Steps	TC-008-A: Submit valid utility bill → verify HTTP 201, UTIL-YYYY-NNNNNN format, DB row, NE-11 email within 2 min TC-008-B: Submit same agreement_id + bill_type + billing_period twice → second attempt returns HTTP 409 ERR-UT-01-003 TC-008-C: ABH approves utility bill → verify NE-12 sent to Finance with bill_id, net_payable, and payment due date TC-008-D: Submit with 10.1 MB file → verify HTTP 422, file not stored on disk TC-008-E: Verify net_payable calculation: bill_amount=10000 + advance_tax=2500 → net_payable must equal 12500.00 in DB

7. System Workflows

Step-by-step workflow tables defining actor, action, system response, status transition, and notification triggered at each step. These are the direct input for sequence diagrams in SDD-001.

7.1 New Rental Request — End-to-End

Step	Actor	Action	System Response	Status After	Notification
1	Area Officer	Fills and submits New Request form	Validates inputs; generates REQ-YYYY-NNNNNN; persists record	PENDING_ABH	NE-01 → ABH (≤2 min)
2a	ABH	Approves request	Updates status; logs approval_history row	PENDING_DIRECTOR	NE-02 → Director (≤2 min)
2b	ABH	Rejects with mandatory remarks	Updates status; logs rejection with remarks; request closed	REJECTED	NE-03 → Area Officer (≤2 min)
3a	Director	Approves request	Updates status; logs approval_history row	PENDING_ADMIN_HO	NE-04 → Admin HO (≤2 min)
3b	Director	Rejects with mandatory remarks	Updates status; logs rejection; request closed	REJECTED	NE-05 → ABH + Area Officer (≤2 min)
4	Admin HO	Reviews documents; inputs/edits details; clicks Complete	Updates status; generates Approval Summary PDF; notifies 2 parties	COMPLETED	NE-06 → Area Admin + Finance simultaneously (≤2 min)
5	System	Auto-generates Approval Summary PDF	PDF stored; URL available in Print Queue (SCR-11)	COMPLETED	—
6	Area Admin	Downloads and prints agreement package from SCR-11	Serves agreement + approval summary as combined PDF package	COMPLETED	—
7	Finance	Processes payment in Oracle Financial using NE-06 notification	Out of scope in i-Property; handled in OF	COMPLETED	—

7.2 Renewal Request Workflow

Step	Actor	Action	System Response	Status After	Notification
1	Admin	Views expiry report (SCR-06); identifies expiring agreements	Report shows agreements expiring within look-ahead window (default 90 days)	N/A	—
2	Area Office	Selects agreement; clicks Initiate Renewal	Pre-populates form with existing data; reserves new reference ID	DRAFT	—

3	ABH/MR/ARH	Edits details, adds remarks, submits renewal	Generates new REQ-YYYY-NNNNNN; cross-references original_request_id	PENDING_ABH	NE-01 → ABH (≤2 min)
4–6	All Approvers	Same chain as New Request Steps 2a–4	Identical to 6.1	→ COMPLETED	As per 6.1
7a	System	If rejected at any stage	Status = CLOSED; no further actions permitted	CLOSED	NE-03 or NE-05 as applicable

7.3 Vacated / Termination Workflow

Step	Actor	Action	System Response	Status After	Notification
1	Area Admin	Opens Vacated Request form; enters termination_date and reason; uploads documents	Validates termination_date not in past; stores form data as DRAFT	DRAFT	—
2	System	On submission	Generates new reference ID; routes to ABH	PENDING_ABH	NE-01 → ABH (≤2 min)
3–4	ABH/Director	Standard approval chain	Same as Steps 2a–3a of 6.1	→ PENDING_ADMIN_HO	NE-02, NE-04
5	System	On Director APPROVE	Sets agreement.payment_stopped=true; enqueues URGENT NE-10 to Finance	PENDING_ADMIN_HO	NE-10 → Finance URGENT (≤2 min)
6	System	Payment scheduler next run	Agreement excluded from monthly_payment_summary generation	—	—
7	Admin HO	Issues vacated notice; uploads scanned copy	Document stored; linked to request; status → COMPLETED	COMPLETED	—

7.4 Utility Bill Payment Workflow

Step	Actor	Action	System Response	Status After	Notification
1	Area Officer	Searches by Agreement ID or Landlord ID on SCR-08	Returns matching records within 2 seconds; shows utility connection list	—	—
2	Area Officer	Selects connection; chooses bill type; fills bill details; uploads	Duplicate check runs; if pass: stores bill; if fail: ERR-UT-01-003	PENDING_ABH	NE-11 → ABH (≤2 min)
3	ABH	Reviews and approves utility bill	Updates status; enqueues NE-12 to Finance	APPROVED	NE-12 → Finance (≤2 min)

4	Finance	Processes utility payment in Oracle Financial	Out of scope in i-Property	—	—

8. API Specifications

All endpoints are prefixed with: https://iproperty.jvo.internal/api/v1

All requests require: Authorization: Bearer {JWT} | Content-Type: application/json (except file upload endpoints which use multipart/form-data)

Full API contracts including all field-level schemas, pagination, and filtering parameters are maintained in IDD-001. This section covers the primary endpoints referenced by the functional requirements.

8.1 Request Management Endpoints

POST /requests — Submit new rental request	
Request Headers	Authorization: Bearer {JWT} Content-Type: multipart/form-data

Request Body	{
	"landlord_id": "LL-000142",
	"property_id": "PROP-000089",
	"area_code": "KHI-EAST",
	"agreement_start_date": "01/05/2025",
	"agreement_end_date": "30/04/2026",
	"rental_amount": 75000.00,
	"tax_rate": 15.00,
	"agreement_type": "New",
	"remarks": "Ground floor office unit"
	}
	+ files[]: document_attachments (multipart)

200 / 201 Response	HTTP 201 Created
	{
	"request_id": "REQ-2025-000142",
	"status": "PENDING_ABH",
	"created_at": "2025-05-01T03:32:11Z",
	"assigned_abh_ids": ["USR-0045"]
	}

Error Responses	400 Bad Request — malformed JSON 401 Unauthorized — missing/expired JWT 403 Forbidden — user area mismatch (ERR-NR-01-003) 422 Unprocessable Entity — field validation failures: { "errors": [{ "field": "agreement_end_date", "code": "ERR-NR-01-005", "message": "End date must be after Start date" }] } 500 Internal Server Error — DB failure (ERR-NR-01-009); stack trace logged server-side only
Notes	Idempotency: duplicate submission of identical payload within 60 seconds returns the existing request_id with HTTP 200 (idempotency key = SHA256 of landlord_id+property_id+start_date+submitted_by) File uploads: stored before DB insert; if DB insert fails, uploaded files are cleaned up (compensating transaction)

PATCH /requests/{request_id}/action — Approve or reject a request at any chain stage

Request Headers	Authorization: Bearer {JWT} Content-Type: application/json

Request Body	{
	"action": "REJECT",
	"remarks": "Rental amount exceeds approved budget ceiling for this area"
	}
	Valid action values: APPROVE | REJECT | COMPLETE (Admin HO only)

200 / 201 Response	HTTP 200 OK
	{
	"request_id": "REQ-2025-000142",
	"new_status": "REJECTED",
	"actioned_by": "USR-0045",
	"actioned_at": "2025-05-02T09:14:33Z"
	}

Error Responses	401 Unauthorized 403 Forbidden — user not permitted to action this request 409 Conflict — request not in expected status (ERR-AP-02-001): { "code": "ERR-AP-02-001", "message": "Current status: PENDING_DIRECTOR", "current_status": "PENDING_DIRECTOR" } 422 Unprocessable — REJECT without remarks (ERR-AP-02-002)
Notes	Optimistic locking: request row includes version integer; PATCH must include current version to prevent concurrent approval conflicts On version mismatch: HTTP 409 with message: 'Record was modified by another user. Please refresh and retry.'

GET /requests/{request_id} — Retrieve full request details including approval history	
Request Headers	Authorization: Bearer {JWT}
Request Body	No body — request_id in path

200 / 201 Response	HTTP 200 OK
	{
	"request_id": "REQ-2025-000142",
	"status": "PENDING_DIRECTOR",
	"landlord": { "id": "LL-000142", "name": "Ahmad Raza" },
	"property": { "id": "PROP-000089", "address": "Shop 5, Block B, Gulshan" },
	"rental_amount": 75000.00,
	"tax_rate": 15.00,
	"approval_history": [
	{ "stage": 1, "role": "ABH", "actor": "Bilal Khan", "action": "APPROVED", "timestamp": "2025-05-02T07:11:00Z", "remarks": null }
	],
	"documents": [{ "doc_type": "AGREEMENT", "url": "https://..." }]
	}

Error Responses	401 Unauthorized 403 Forbidden — user not permitted to view this request 404 Not Found
Notes	Response filtered by role: Finance can see agreement financials but not landlord bank details; Area Officer cannot see other areas' requests

8.2 Utility Bill Endpoints

POST /utility-bills — Submit a utility bill for approval	
Request Headers	Authorization: Bearer {JWT} Content-Type: multipart/form-data

Request Body	{
	"agreement_id": "AGR-2025-000031",
	"bill_type": "ELECTRICITY",
	"billing_period": "04/2025",
	"units_consumed": 340.50,
	"bill_amount": 10200.00,
	"advance_tax": 1530.00
	}
	+ file: bill_document (multipart, required)

200 / 201 Response	HTTP 201 Created
	{
	"utility_bill_id": "UTIL-2025-000089",
	"status": "PENDING_ABH",
	"net_payable": 11730.00,
	"created_at": "2025-04-28T11:05:44Z"
	}

Error Responses	409 Conflict — duplicate bill (ERR-UT-01-003): { "code": "ERR-UT-01-003", "existing_bill_id": "UTIL-2025-000061", "message": "Duplicate bill for ELECTRICITY / 04/2025" } 422 Unprocessable — field validation failures 401, 403, 500 — same pattern as /requests
Notes	net_payable = bill_amount + advance_tax; calculated server-side; client-submitted net_payable (if any) is ignored

8.3 Oracle Financial Integration Endpoints

The following outbound calls are made by i-Property to Oracle Financial. Full payload schemas, field mappings, authentication mechanism (API Key / OAuth), and retry policy are specified in IDD-001 S3.

POST [OF] /payment-requests — Transmit approved monthly payment batch to Oracle Financial	
Request Headers	Authorization: ApiKey {OF_API_KEY} (stored in server-side env var — never in client) Content-Type: application/json X-Source-System: I-PROPERTY X-Idempotency-Key: {monthly_summary_id}
Request Body	{ "batch_reference": "PAY-BATCH-2025-04", "generated_by": "I-PROPERTY", "approved_at": "2025-04-30T06:00:00Z", "payments": [{ "landlord_id": "LL-000142", "agreement_id": "AGR-2025-000031", "period": "04/2025", "gross_amount": 75000.00,

	"tax_deduction": 11250.00, "net_payable": 63750.00, "bank_account": "36ALFA0021000005854802", "payment_reference": "IPROP-REQ-2025-000142-04-2025" }] }

200 / 201 Response	HTTP 201 Created (from OF)
	{
	"of_batch_id": "OF-BATCH-20250430-0042",
	"status": "QUEUED",
	"payment_count": 1
	}
	i-Property: stores of_batch_id against monthly_payment_summary record

Error Responses	OF 400 — schema validation failure: log error; System Admin alerted; Finance notified via NE with error details OF 401/403 — auth failure: System Admin alerted; retry NOT automatic (requires credential check) OF 5xx — server error: retry 3 times at 10-minute intervals with exponential backoff; if still failing after 3 retries: System Admin + Finance alerted with batch_reference Network timeout (>30 seconds): treat as 5xx; same retry policy
Notes	Idempotency: X-Idempotency-Key prevents duplicate payment batches if i-Property retries bank_account transmitted encrypted via TLS; not logged in application logs Full field mapping table in IDD-001 §3.1 Table 2

POST [OF] /expense-accruals — Push expense accrual entries for ledger recording	
Request Headers	Authorization: ApiKey {OF_API_KEY} Content-Type: application/json

Request Body	{
	"entries": [
	{
	"entry_type": "ACCRUAL",
	"agreement_id": "AGR-2025-000031",
	"period": "04/2025",
	"amount": 75000.00,
	"cost_centre": "KHI-EAST",
	"ledger_account": "7100-RENT"
	}
	]
	}

200 / 201 Response	HTTP 201 from OF with entry reference IDs i-Property updates request record with of_accrual_ref
Error Responses	Same retry and alert policy as payment-requests endpoint
Notes	Triggered when Admin HO marks request as COMPLETED Ledger account codes to be confirmed with Finance team and documented in IDD-001 §3.3

9. UI Screen Inventory & Requirements

Detailed wireframes, component specifications, and interaction design are maintained in UI-001. This section defines the functional obligations of each screen — the contract between this SRS and UI-001.

Screen ID	Screen Name	Roles	Key Functional Obligations	UI-001 Ref
SCR-01	Login / SSO	All	Redirect to SSO; on success receive JWT; on failure show lockout state after 5 attempts	UI-001 §2.1
SCR-02	Dashboard	All (role-filtered)	Show: pending approvals count badge, expiry alert count, quick-action links. Role-specific widgets (Finance sees payment reports; ABH sees approval queue). All counts reflect live DB state.	UI-001 §2.2
SCR-03	New Request Form	Area Officer	Multi-section form with Landlord lookup (typeahead), Property lookup (filtered by landlord), Agreement details, Document upload. All mandatory fields marked *. Inline validation on blur. Discard prompt on navigation (FR-NR-05).	UI-001 §3.1
SCR-04	Approval Queue	ABH, Director, Admin HO	Paginated list of pending requests. Columns: Reference ID, Type, Area, Landlord Name, Submitted Date, Days Pending (highlighted red if SLA breached). Click row → SCR-05. Bulk approve not permitted.	UI-001 §3.2
SCR-05	Request Detail	All roles	Full request data, editable fields for Admin HO only. Approval history timeline (visual). Document viewer (inline PDF/image). Approve/Reject/Complete action buttons with confirmation modal. Remarks field shown on Reject.	UI-001 §3.3
SCR-06	Renewal / Expiry Report	Admin, Area Office, ABH	Table of agreements filtered by expiry look-ahead (configurable, default 90 days). Columns: Agreement ID, Landlord, Area, Expiry Date, Days Remaining (colour-coded: red <30, amber 30-60, green 60-90). Initiate Renewal button per row.	UI-001 §3.4
SCR-07	Vacated Request Form	Area Admin	Captures: termination_date (date picker, not in past), reason (text, min 10 chars), document upload (required). Same discard prompt as SCR-03.	UI-001 §3.5
SCR-08	Utility Bills Screen	Area Officer	Search bar (by Agreement ID or Landlord ID), results list with connection type. Bill entry form: bill type dropdown, period picker (MM/YYYY), numeric inputs for amounts. Inline duplicate warning before submit.	UI-001 §4.1
SCR-09	Monthly Payment Summary	ABH/GM, Finance	ABH view: editable grid with stop-payment toggle per row (requires reason). Approve button (confirmation modal). Finance view: read-only; download Excel/PDF buttons. Stopped rows visually distinguished (strikethrough + amber highlight).	UI-001 §5.1
SCR-10	Reports Module	Admin, ABH, Finance	Report type selector. Filter panel (area, date range, status, landlord). Export to Excel and PDF. All filters apply before export. Table	UI-001 §6.1

			preview before export (max 100 rows shown; full data in export).	
SCR-11	Print Queue	Area Admin	Lists completed requests with Agreement document + Approval Summary PDF. Download Combined Package button (merges documents into single PDF for printing).	UI-001 §6.2
SCR-12	User Management	System Admin	Create/edit/deactivate users. Role assignment (multi-role not permitted). Area mapping. Password reset trigger. Audit log access button.	UI-001 §7.1
SCR-13	Notification Config	System Admin	Edit HTML templates for all 14 notification events. Configure SLA breach thresholds per approval stage. Test Send button (sends to System Admin email). Template variables listed per template.	UI-001 §7.2

9.1 Global UI Requirements

- All mandatory fields: red asterisk (*) label suffix; unfilled mandatory field on submit highlights border red with message below
- All API error responses: display user-facing message from error.message field; never expose error codes, stack traces, or internal IDs in UI
- Loading states: any action that calls an API must show a loading spinner on the action button; button disabled during request to prevent double-submission
- Session timeout: 28 minutes after last activity → modal: 'Your session will expire in 2 minutes. Click to stay logged in.' At 30 min: auto-logout, redirect to SCR-01, flash message: 'Your session has expired. Please log in again.'
- All tables: default sort by created_at DESC; column headers clickable for sort toggle; sort state preserved on page refresh (stored in URL params)
- File upload: show filename, file size, and remove (X) button after selection; progress bar during upload; error shown inline if type/size validation fails
- Confirmation modals for destructive/irreversible actions (Approve, Reject, Stop Payment, Discard): modal title, explanation sentence, and two buttons — primary action (coloured) + cancel (outline)

10. Field Definitions

Field types and constraints defined here represent the application-layer contract. Database column types (including indexes and foreign keys) are elaborated in SDD-001 §4. Error messages shown here are the exact strings the UI must display.

10.1 New Rental Request — Agreement Fields

Field Name	DB Type	Max Length	Mandatory	Validation Rule	Error Message	Notes
reference_id	VARCHAR	20	Auto	System-generated; format: REQ-YYYY-NNNNNN; DB unique constraint	—	Read-only; generated post-submit

landlord_id	VARCHAR	20	Yes	Must exist in landlord_profile with status=ACTIVE	Landlord ID does not exist or is inactive	Typeahead lookup on SCR-03
property_id	VARCHAR	20	Yes	Must exist in property_profile linked to landlord_id with status=ACTIVE	Property not associated with selected Landlord	Filtered by landlord_id selection
area_code	VARCHAR	20	Yes	Must exist in areas table; must be in user's allowed_areas	You do not have permission for this area	Dropdown; filtered by user's areas
agreement_type	ENUM	—	Yes	Values: New, Renewal, Transfer	Please select an agreement type	
start_date	DATE	—	Yes	Not in past (today or future); format DD/MM/YYYY	Start date cannot be in the past	Date picker; past dates disabled
end_date	DATE	—	Yes	Strictly > start_date	End date must be after Start date	Validates after start_date selected
rental_amount	DECIMAL	15,2	Yes	Range: 1.00 – 99,999,999.99; no negative; numeric only	Amount must be between 1.00 and 99,999,999.99	Formatted with thousand separators in UI
tax_rate	DECIMAL	5,2	Yes	Range: 0.00 – 100.00; default: loaded from system tax config	Tax rate must be between 0 and 100	Pre-filled; editable
escalation_clause	TEXT	500	No	Free text; max 500 characters	Maximum 500 characters	Character counter shown in UI
remarks	TEXT	500	No	General remarks; mandatory on REJECT (min 1 char, enforced at approval step not here)	Remarks are required to reject	Shown/required only on Reject action
document_attachments	FILE[]	10 MB	No	Types: PDF, JPG, PNG; max 10 MB per file; max 5 files per submission	Only PDF, JPG, PNG permitted / Exceeds 10 MB	Multiple upload; each file validated
status	ENUM	—	Auto	Values: DRAFT, PENDING_ABH, PENDING_DIRECTOR, PENDING_ADMIN_HO, COMPLETED, REJECTED, CLOSED	—	System-controlled; read-only in UI

created_by	VARCHAR	36	Auto	user_id from JWT; cannot be manually set	—	From session context
created_at	TIMESTAMP	—	Auto	UTC timestamp set by DB on INSERT; displayed in PST in UI	—	
version	INTEGER	—	Auto	Optimistic lock counter; incremented on each UPDATE; default 1	—	Sent in PATCH requests to prevent conflicts

10.2 Utility Bill Fields

Field Name	DB Type	Max Length	Mandatory	Validation Rule	Error Message	Notes
utility_bill_id	VARCHAR	20	Auto	Format: UTIL-YYYY-NNNNNN; DB unique constraint	—	Auto-generated
agreement_id	VARCHAR	20	Yes	FK to agreements; agreement.status must = ACTIVE	Agreement is not active or does not exist	Lookup with search
bill_type	ENUM	—	Yes	Values: ELECTRICITY, GAS, WATER, OTHER	Please select a bill type	Dropdown
billing_period	VARCHAR	7	Yes	Format: MM/YYYY; not in future; validated as: month 1-12, year >= current year - 5	Billing period cannot be in the future	Month/Year picker
units_consumed	DECIMAL	10,2	Yes	Min: 0; no negative; numeric only	Units consumed cannot be negative	
bill_amount	DECIMAL	15,2	Yes	Min: 0.01; no negative	Bill amount must be greater than zero	
advance_tax	DECIMAL	15,2	No	Min: 0; default: 0; no negative	Advance tax cannot be negative	
net_payable	DECIMAL	15,2	Calc	Server-calculated: bill_amount + advance_tax; any client-submitted value ignored	—	Displayed read-only after submit

bill_document	FILE	10 MB	Yes	Types: PDF, JPG, PNG; max 10 MB; exactly 1 file required	Please upload the bill document	
status	ENUM	—	Auto	Values: PENDING_ABH, APPROVED, REJECTED	—	System-controlled

10.3 Landlord Profile Fields

Field Name	DB Type	Max Length	Mandatory	Validation Rule	Error Message	Notes
landlord_id	VARCHAR	20	Auto	Format: LL-NNNNNN; DB unique constraint	—	Auto-generated on profile creation
full_name	VARCHAR	100	Yes	Min 2 chars; letters, spaces, hyphens, periods only; no numbers	Name must be at least 2 characters	
state_id_dl_number	VARCHAR	20	Yes	Alphanumeric (state-specific; no single standard format) DB Constraint: Unique per user (recommended)	Invalid ID format (state-specific validation optional)	Input: No strict mask enforced; allow flexible entry due to varying formats
contact_number	VARCHAR	15	Yes	U.S. phone number (mobile or landline) Pattern: (XXX) XXX-XXXX or XXX-XXX-XXXX (10 digits)	Invalid contact number format	Allow flexible formats; normalize to 10-digit numeric for storage
email_address	VARCHAR	100	No	RFC 5322 email format; if provided, must be unique in landlord_profile table	Invalid email address format	
mailing_address	TEXT	255	Yes	Min 10 chars; free text	Please enter a complete mailing address	
bank_name	VARCHAR	100	Yes	Min 2 chars; free text	Please enter the bank name	
bank_account_no	VARCHAR	30	Yes	Numeric only; min 10 digits; stored AES-256 encrypted at rest; displayed	Invalid bank account number	Never logged; masked in all API responses

				masked (last 4 visible) in UI		
iban	VARCHAR	34	No	Format: + 2 check digits + 4 bank code + 16 account digits; IBAN checksum validated server-side	Invalid IBAN	
tax_registration	ENUM	—	Yes	Values: REGISTERED, UNREGISTERED, EXEMPT	Please select tax registration status	
status	ENUM	—	Yes	Values: ACTIVE, INACTIVE; default: ACTIVE	—	INACTIVE blocks new agreements

11. Non-Functional Requirements

Each NFR includes the specification, the precise verification method, and the document that owns the implementation detail.

NFR-ID	Category	Specification	Verification Method	Owner Doc	Priority
NFR-01	Performance	Average API response time ≤ 500ms at 50 concurrent users Average page load (TTI) ≤ 3 seconds at 50 concurrent users No degradation threshold: system must handle 100 concurrent users with ≤ 20% response time increase Report export (10,000 rows): ≤ 30 seconds	JMeter load test: 100 virtual users, 30-min sustained ramp Measure P95 response time per endpoint Test to be documented in TP-001 §8	DP-001	**High**
NFR-02	Availability	99.5% uptime during business hours 08:00–20:00 PST Mon–Sat RTO (Recovery Time Objective): ≤ 1 business day RPO (Recovery Point Objective): ≤ 24 hours Planned maintenance window: Sundays 02:00–06:00 PST	Uptime monitoring via Nagios/Zabbix (to be configured — DP-001 §4) Monthly uptime report reviewed by IT Head	DP-001	**High**
NFR-03	Authentication	SSO/LDAP via OAuth 2.0 or SAML 2.0 JWT expiry: 8 hours; refresh token: 24 hours	Penetration test: attempt API calls without/with expired JWT — verify 401	SAD-001	**Critical**

		Session inactivity timeout: 30 minutes with 2-minute warning Account lockout: 5 consecutive failures; unlock by Admin or 30-min auto-reset All API endpoints: 401 returned within 50ms for invalid/missing token	Functional test: trigger lockout after 5 failures — verify account locked Session timeout test: leave session idle 32 minutes — verify auto-logout and redirect		
NFR-04	Authorisation (RBAC)	RBAC enforced at API endpoint level (not only UI level) Area Officer from Area A cannot access Area B data Finance role cannot POST to /requests or /utility-bills All permission checks logged in audit_log	Role-based test matrix: for each of 7 roles, attempt access to each of 13 screen endpoints — verify correct allow/deny Cross-area test: Area Officer USR-A attempts GET /requests?area=AREA-B — verify 403 Test cases documented in TP-001 §5	SAD-001	**Critical**
NFR-05	Data Security	TLS 1.2 minimum for all data in transit; TLS 1.3 preferred AES-256 encryption at rest for: bank_account_no, iban bank_account_no: never returned in full in any API response (masked: last 4 digits) No sensitive field values in server logs or client-side storage HTTPS enforced; HTTP requests redirected to HTTPS	SSL certificate audit: verify TLS version and cipher suite DB column audit: verify encrypted columns using SELECT on encrypted_bank_account API response audit: verify bank_account_no masked in all GET /requests responses Log audit: verify no bank account data in application logs	SAD-001	**Critical**
NFR-06	Audit Trail	All INSERT, UPDATE, DELETE on core tables: logged with user_id, ip_address, timestamp_utc, old_value(JSON), new_value(JSON) audit_log table: no UPDATE or DELETE permitted at DB level (enforced by DB role) Audit logs retained: 3 years active, 7 years archived (cold storage) System Admin can search audit_log by user, date range, table, and record_id	Create a request, then approve it — verify audit_log contains both INSERT and UPDATE entries with correct before/after JSON Attempt DELETE on audit_log as application DB user — verify permission denied View audit log via SCR-12 — verify search and filter functions	SAD-001	**High**

NFR-07	Browser Compatibility	Supported: Chrome (current-2), Firefox (current-2), Edge (current-2) Minimum legacy: Internet Explorer 11 (degraded but functional; no graceful-degradation guarantee for IE11 only) Mobile browsers: not in scope for Phase 1	Cross-browser automated regression suite (Selenium/Playwright) covering all 13 screens Run against Chrome, Firefox, Edge on Windows 10 and 11 IE11 smoke test: submit request, approve, export report — no blocking errors	TP-001	**Medium**
NFR-08	Usability	Core workflows (New Request, Renewal, Utility Bill): completable by trained user in ≤ 10 screen interactions All form fields: inline validation message within 300ms of blur event All user-facing error messages: plain language; no technical codes or stack traces Accessibility: WCAG 2.1 AA target (to be confirmed during UI-001 design phase)	Walkthrough test: count screen interactions for New Request happy path — must be ≤ 10 Error message review: QA to verify all 422 error messages in TP-001 §7 are plain language Blur validation timing test: measure validation message appearance delay	UI-001	**Medium**
NFR-09	Scalability	Architecture must support 150 concurrent users (50% growth) without code changes Horizontal scaling via stateless application tier (no server-side session state — JWT-based) DB connection pooling configured (min 10, max 50 connections)	Architecture review in SAD-001: verify stateless design Load test at 150 virtual users: verify ≤ 50% degradation from 100-user baseline	SAD-001, DP-001	**Medium**
NFR-10	Data Backup	Full database backup: daily at 02:00 PST Incremental backup: every 6 hours Backup retention: 90 days Restore tested: quarterly drill with documented RTO verification Backup storage: offsite / separate from primary DB server	Review backup schedule configuration in DP-001 Quarterly restore drill: restore latest backup to test environment; verify data integrity; document time taken; compare against RTO ≤ 1 business day	DP-001	**High**

12. Notification Event Specifications

All notifications are delivered asynchronously via the notification job queue (INT-04 SMTP relay). On first delivery failure, retry 3 times at 5-minute intervals. If all retries fail: log to notification_failures table; alert System Admin; do not block the triggering transaction.

NE-ID	Trigger	Recipients	Subject Line	Required Body Fields	Delivery SLA	Priority
NE-01	New request submitted (POST /requests success)	ABH mapped to request's area_code	[Action Required] New Rental Request: {reference_id}	reference_id landlord_name property_address area_name submission_date (PST) direct_approval_url	≤ 2 min	**High**
NE-02	ABH approves → status = PENDING_DIRECTOR	Director Office users	[Action Required] Rental Request Approved by ABH: {reference_id}	reference_id landlord_name abh_name approval_date direct_approval_url	≤ 2 min	**High**
NE-03	ABH rejects → status = REJECTED	Area Officer who submitted	[Update] Rental Request Rejected: {reference_id}	reference_id landlord_name rejected_by rejection_date rejection_remarks new_request_url	≤ 2 min	**High**
NE-04	Director approves → status = PENDING_ADMIN_HO	Admin Head Office users	[Action Required] Request Approved by Director: {reference_id}	reference_id landlord_name director_name approval_date direct_review_url	≤ 2 min	**High**
NE-05	Director rejects → status = REJECTED	ABH + originating Area Officer	[Update] Request Rejected by Director: {reference_id}	reference_id landlord_name rejected_by rejection_remarks	≤ 2 min	**High**
NE-06	Admin HO completes → status = COMPLETED	Area Admin AND Finance Dept. (simultaneous)	[Action Required] Agreement Ready: {reference_id}	reference_id agreement_id landlord_name property_address rental_amount tax_rate effective_date approval_summary_url	≤ 2 min	**High**
NE-07	Monthly payment summary generated	All ABH/GM users	[Action Required] Monthly Payment	period total_agreements total_gross_amount total_net_payable	By 08:05 PST generation day	**High**

			Summary Ready: {period}	summary_review_url		
NE-08	ABH approves payment summary	Finance Dept.	[Action] Approved Payment Summary: {period}	period approved_by approval_date payment_count total_net_payable download_url (Excel + PDF)	≤ 2 min	**High**
NE-09	Weekly scheduled job (Monday 08:00 PST)	Admin + ABH	[Weekly Report] Agreements Expiring Within {n} Days	report_period expiry_count agreements_list (id, landlord, expiry_date, days_remaining) renewal_initiation_url	By 08:10 PST Monday	**Medium**
NE-10	Director approves vacated → payment_stopped=true	Finance Dept. (URGENT)	[URGENT — Action Required] Stop Payment: Agreement {agreement_id}	agreement_id landlord_id landlord_name bank_account (masked) termination_date last_payment_date action_required: 'Stop all future payments from the date above'	≤ 2 min	**Critical**
NE-11	Utility bill submitted (POST /utility-bills success)	ABH for agreement's area	[Action Required] Utility Bill Pending Approval: {utility_bill_id}	utility_bill_id agreement_id landlord_name bill_type billing_period net_payable bill_document_url approval_url	≤ 2 min	**High**
NE-12	ABH approves utility bill	Finance Dept.	[Action] Utility Bill Approved for Payment: {utility_bill_id}	utility_bill_id agreement_id landlord_name bill_type billing_period net_payable payment_due_date	≤ 2 min	**High**
NE-13	Stop-payment flag set on landlord in summary	Finance Dept. + ABH	[Alert] Payment Stopped: Landlord {landlord_id}	landlord_id landlord_name period stop_reason stopped_by	≤ 2 min	**High**

NE-14	Approval SLA breached (pending > configured threshold per stage)	System Admin + relevant manager	[SLA Breach] Approval Pending: {reference_id}	reference_id current_stage assigned_to submitted_at hours_pending sla_threshold_hours	≤ 5 min of breach	**Medium**

NE-ID	Trigger	Recipients	Subject	Key Body Fields	SLA	Priority
NE-01	POST /requests success	ABH (area mapped)	[Action Required] New Request: {ref_id}	ref_id, landlord_name, property, area, date, approval_url	≤2 min	**High**
NE-02	ABH APPROVE	Director Office	[Action Required] ABH Approved: {ref_id}	ref_id, landlord, abh_name, date, approval_url	≤2 min	**High**
NE-03	ABH REJECT	Area Officer	[Rejected] Request: {ref_id}	ref_id, rejected_by, date, remarks, resubmit_url	≤2 min	**High**
NE-04	Director APPROVE	Admin Head Office	[Action Required] Director Approved: {ref_id}	ref_id, landlord, director_name, date, review_url	≤2 min	**High**
NE-05	Director REJECT	ABH + Area Officer	[Rejected] Director Rejected: {ref_id}	ref_id, rejected_by, remarks	≤2 min	**High**
NE-06	Admin HO COMPLETE	Area Admin + Finance (both)	[Ready] Agreement Complete: {ref_id}	ref_id, agreement_id, landlord, rental_amount, tax_rate, effective_date, pdf_url	≤2 min	**High**
NE-07	Payment summary auto-generated	All ABH/GM	[Action Required] Monthly Summary Ready: {period}	period, total_agreements, gross, net, review_url	08:05 PST generation day	**High**
NE-08	Summary APPROVED by ABH	Finance Dept.	[Action] Approved Summary: {period}	period, approved_by, count, total_net, download_url	≤2 min	**High**
NE-09	Weekly scheduler (Mon 08:00)	Admin + ABH	[Weekly] Expiring Agreements: {n} within {days}d	expiry_count, agreement list, renewal_url	By 08:10 Mon	**Medium**
NE-10	Director APPROVE vacated (URGENT)	Finance Dept.	[URGENT] Stop Payment: Agreement {agr_id}	agreement_id, landlord_id, bank (masked), termination_date, last_payment_date	≤2 min	**Critical**

NE-11	POST /utility-bills success	ABH (area mapped)	[Action Required] Utility Bill Pending: {bill_id}	bill_id, agreement_id, landlord, bill_type, period, net_payable, approval_url	≤2 min	**High**
NE-12	ABH approves utility bill	Finance Dept.	[Action] Utility Bill Approved: {bill_id}	bill_id, agreement_id, landlord, bill_type, period, net_payable, due_date	≤2 min	**High**
NE-13	Stop-payment flag set on landlord	Finance + ABH	[Alert] Payment Stopped: Landlord {ll_id}	landlord_id, landlord_name, period, stop_reason, stopped_by	≤2 min	**High**
NE-14	Approval SLA breached	System Admin + Manager	[SLA Breach] Pending: {ref_id}	ref_id, current_stage, assigned_to, hours_pending, sla_threshold	≤5 min of breach	**Medium**

13. Requirements Traceability Matrix

Full RTM maintained in TRM-001. Preview below maps each BRD requirement to the SRS FR, processing logic section, UI screen, API endpoint, and test case reference.

BRD Ref	Requirement Summary	SRS FR	SRS §	API Endpoint	UI Screen	TC Ref
FR-NR-01	Auto-generate reference ID	FR-NR-01	5.1	POST /requests	SCR-03	TC-001-A/B/D
FR-NR-02	Email ABH on submission	FR-NR-01	5.1	Async NE-01	—	TC-001-A
FR-NR-03	Rejection requires remarks	FR-NR-02	5.2	PATCH /requests/{id}/action	SCR-04/05	TC-002-B
FR-NR-04	Full ABH→Director→Admin HO routing	FR-NR-02 /03	5.2	PATCH /requests/{id}/action	SCR-04/05	TC-002-A, TC-003-A
FR-NR-05	Area Admin + Finance notified on complete	FR-NR-03	5.2	Async NE-06	—	TC-003-B/C
FR-NR-08	Approval summary PDF generation	FR-NR-04	5.3	Internal PDF engine	SCR-11	TC-004-A/B/C
FR-NR-12	Discard unsaved changes	FR-NR-05	5.4	Browser beforeunload + router guard	SCR-03/07/08	TC-005-A to E
FR-NR-09	Monthly payment summary	FR-NR-06	5.5	POST /payment-summary /generate	SCR-09	TC-006-A to E
FR-VT-02	Stop payment on vacated approval	FR-VT-01	5.6	PATCH /requests/{id}/action	SCR-05	TC-007-A to D

FR-UT-01 to 04	Utility bill with duplicate check	FR-UT-01	5.7	POST /utility-bills	SCR-08	TC-008-A to E
NFR-01 to 10	All non-functional requirements	NFR-01 to 10	§10	N/A — architectural	N/A	TP-001 §§5–9
Integration OF	Oracle Financial payment batch	FR-NR-06	§7	POST [OF]/payment-requests	—	IDD-001 integration tests
Integration OF	Oracle Financial expense accruals	FR-NR-03	§7	POST [OF]/expense-accruals	—	IDD-001 integration tests

14. Document Approval & Sign-Off

Changes after sign-off require a formal Change Request submitted to IT-PMO. The version number will be incremented accordingly.

Name & Role	Department	Date	Signature
Business Analyst, IT-PMO	IT — PMO		
Head of Enterprise Architecture	IT — PMO		
Manager Real Estate	Admin		
Senior Manager Finance	Finance		
Head Business Systems	IT		
Head of Internal Audit	Audit		
Head of Legal	Legal		